

Cursors Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE, ENROLLMENT.

AIM: To declare and use cursors inside a MySQL stored procedure to process query results row by row.

A cursor needs 4 things in this order: DECLARE cursor, DECLARE a CONTINUE HANDLER for NOT FOUND, OPEN, then a loop that FETCHes until the handler flag is set, then CLOSE.

-- 1. Basic cursor: loop through all faculty names

```
DELIMITER //
CREATE PROCEDURE ListFacultyNames()
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE fname VARCHAR(15);
    DECLARE facCursor CURSOR FOR SELECT FacName FROM FACULTY;
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    OPEN facCursor;
    read_loop: LOOP
        FETCH facCursor INTO fname;
        IF done = 1 THEN
            LEAVE read_loop;
        END IF;
        SELECT fname AS FacultyName;
    END LOOP;
    CLOSE facCursor;
END //
DELIMITER ;

CALL ListFacultyNames();
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL ListFacultyNames();
+-----+
| FacultyName |
+-----+
| Ramesh      |
+-----+
1 row in set (0.02 sec)

+-----+
| FacultyName |
+-----+
| Mohan       |
+-----+
1 row in set (0.03 sec)

+-----+
| FacultyName |
+-----+
| Sunitha     |
+-----+
1 row in set (0.03 sec)

+-----+
| FacultyName |
+-----+
| Kavya       |
+-----+
1 row in set (0.03 sec)

+-----+
| FacultyName |
+-----+
| Arjun       |
+-----+
1 row in set (0.03 sec)

Query OK, 0 rows affected (0.03 sec)
```

-- 2. Cursor that accumulates a total (running sum)

```
DELIMITER //
CREATE PROCEDURE TotalFacultyAge(OUT totalAge INT)
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE thisAge INT;
    DECLARE ageCursor CURSOR FOR SELECT Age FROM FACULTY;
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    SET totalAge = 0;
    OPEN ageCursor;
    sum_loop: LOOP
        FETCH ageCursor INTO thisAge;
        IF done = 1 THEN
            LEAVE sum_loop;
        END IF;
        SET totalAge = totalAge + thisAge;
    END LOOP;
    CLOSE ageCursor;
END //
DELIMITER ;

CALL TotalFacultyAge(@total);
SELECT @total AS TotalAge;
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE TotalFacultyAge(OUT totalAge INT)
-> BEGIN
->     DECLARE done INT DEFAULT 0;
->     DECLARE thisAge INT;
->     DECLARE ageCursor CURSOR FOR SELECT Age FROM FACULTY;
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     SET totalAge = 0;
->     OPEN ageCursor;
->     sum_loop: LOOP
->         FETCH ageCursor INTO thisAge;
->         IF done = 1 THEN
->             LEAVE sum_loop;
->         END IF;
->         SET totalAge = totalAge + thisAge;
->     END LOOP;
->     CLOSE ageCursor;
-> END //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL TotalFacultyAge(@total);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @total AS TotalAge;
+-----+
| TotalAge |
+-----+
|      272 |
+-----+
1 row in set (0.00 sec)
```

-- 3. Cursor with a WHERE-filtered query

```
DELIMITER //
CREATE PROCEDURE ListCSEFaculty()
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE fname VARCHAR(15);
    DECLARE cseCursor CURSOR FOR
        SELECT FacName FROM FACULTY WHERE Dept = 'CSE';
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    OPEN cseCursor;
cse_loop: LOOP
    FETCH cseCursor INTO fname;
    IF done = 1 THEN
        LEAVE cse_loop;
    END IF;
    SELECT fname AS CSE_Faculty;
END LOOP;
CLOSE cseCursor;
END //
DELIMITER ;

CALL ListCSEFaculty();
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL ListCSEFaculty();
+-----+
| CSE_Faculty |
+-----+
| Ramesh      |
+-----+
1 row in set (0.00 sec)

+-----+
| CSE_Faculty |
+-----+
| Mohan       |
+-----+
1 row in set (0.00 sec)

+-----+
| CSE_Faculty |
+-----+
| Arjun       |
+-----+
1 row in set (0.01 sec)
```

-- 4. Cursor that updates rows one at a time

```
DELIMITER //
CREATE PROCEDURE GiveSeniorRaise()
BEGIN
    DECLARE done INT DEFAULT 0;
    DECLARE fno VARCHAR(4);
    DECLARE seniorCursor CURSOR FOR
        SELECT FacNo FROM FACULTY WHERE Age >= 55;
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

    OPEN seniorCursor;
    update_loop: LOOP
        FETCH seniorCursor INTO fno;
        IF done = 1 THEN
            LEAVE update_loop;
        END IF;
        UPDATE FACULTY SET Age = Age + 0 WHERE FacNo = fno; -- placeholder update
    END LOOP;
    CLOSE seniorCursor;
END //
DELIMITER ;

CALL GiveSeniorRaise();
SELECT * FROM FACULTY WHERE Age >= 55;
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE GiveSeniorRaise()
-> BEGIN
->     DECLARE done INT DEFAULT 0;
->     DECLARE fno VARCHAR(4);
->     DECLARE seniorCursor CURSOR FOR
->         SELECT FacNo FROM FACULTY WHERE Age >= 55;
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN seniorCursor;
->     update_loop: LOOP
->         FETCH seniorCursor INTO fno;
->         IF done = 1 THEN
->             LEAVE update_loop;
->         END IF;
->         UPDATE FACULTY SET Age = Age + 0 WHERE FacNo = fno; -- placeholder update
->     END LOOP;
->     CLOSE seniorCursor;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL GiveSeniorRaise();
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM FACULTY WHERE Age >= 55;
+-----+-----+-----+-----+-----+-----+-----+-----+
| FacNo | FacName | Gender | DOB       | DOJ       | MobileNo | Dept | Age |
+-----+-----+-----+-----+-----+-----+-----+-----+
| F002  | Mohan   | M      | 1965-11-02 | 1998-07-15 | 9840012346 | CSE | 59 |
| F004  | Kavya   | F      | 1958-01-10 | 1990-06-10 | 9840012348 | IT  | 68 |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

